

Q. 1 (Tiger-ants). Define $\{B_t^{(1)}\}_{t \geq 0}$ and $\{B_t^{(2)}\}_{t \geq 0}$ as the two independent standard Brownian motions that represent the two ants' locations.

a. For $t > s$, we have

$$\begin{aligned}
 \mathbb{E} \left[\left(B_t^{(1)} \right)^2 \mid B_s^{(1)} = x \right] &= \mathbb{E} \left[\left(B_s^{(1)} + B_t^{(1)} - B_s^{(1)} \right)^2 \mid B_s^{(1)} = x \right] \\
 &= \mathbb{E} \left[\left(B_s^{(1)} \right)^2 \mid B_s^{(1)} = x \right] + 2\mathbb{E} \left[B_s^{(1)} \left(B_t^{(1)} - B_s^{(1)} \right) \mid B_s^{(1)} = x \right] \\
 &\quad + \mathbb{E} \left[\left(B_t^{(1)} - B_s^{(1)} \right)^2 \mid B_s^{(1)} = x \right] \\
 &= x^2 + 2x\mathbb{E} \left[B_t^{(1)} - B_s^{(1)} \mid B_s^{(1)} = x \right] + \mathbb{E} \left[\left(B_t^{(1)} - B_s^{(1)} \right)^2 \mid B_s^{(1)} = x \right] \\
 &\stackrel{(i)}{=} x^2 + 2x\mathbb{E} \left[B_t^{(1)} - B_s^{(1)} \right] + \mathbb{E} \left[\left(B_t^{(1)} - B_s^{(1)} \right)^2 \right] \\
 &\stackrel{(ii)}{=} x^2 + 2x \cdot 0 + t - s \\
 &= x^2 + t - s.
 \end{aligned}$$

Here (i) uses the independent increments property that $B_t^{(1)} - B_s^{(1)} \perp B_s^{(1)}$, and (ii) comes from the Gaussian increments property $B_t^{(1)} - B_s^{(1)} \sim N(0, t - s)$.

b. We need to calculate the probability of the event $\{|B_t^{(1)} - B_t^{(2)}| \leq r\}$. Note that

$$B_t^{(1)} \sim N(0, t), \quad B_t^{(2)} \sim N(0, t), \quad B_t^{(1)} \perp B_t^{(2)} \implies B_t^{(1)} - B_t^{(2)} \sim N(0, 2t).$$

Therefore we have

$$\mathbb{P} \left(|B_t^{(1)} - B_t^{(2)}| \leq r \right) = \int_{-r}^r \frac{1}{\sqrt{4\pi t}} \exp \left(-\frac{x^2}{4t} \right) dx = \sqrt{\frac{1}{\pi t}} \int_0^r \exp \left(-\frac{x^2}{4t} \right) dx.$$

Grading comments for Q. 1

- a.
 - Many students are not using stationary increments property of Brownian motion correctly. For example, a common mistake is to use $B_t - B_s = B_{t-s}$, which is not correct. We only know that the random variables $B_t - B_s$ and B_{t-s} have the same distribution, which is $N(0, t - s)$. This doesn't mean that the two random variables are identical.
 - A few students are not using independent increments property of Brownian motion correctly. For example, a common mistake is as follows:

$$\mathbb{E} \left[\left(B_t^{(1)} - B_s^{(1)} \right)^2 \mid B_s^{(1)} = x \right] = \mathbb{E} \left[\left(B_t^{(1)} - x \right)^2 \mid B_s^{(1)} = x \right] = \mathbb{E} \left[\left(B_t^{(1)} - x \right)^2 \right],$$

where they claim that the last step holds because of the independent increments property. The last step is not correct, since we only know that the random variables $B_t^{(1)} - B_s^{(1)}$ and $B_s^{(1)}$ are independent. The last step mistakenly treat $B_t^{(1)} - x$ and $B_s^{(1)}$ as two independent random variables, which does not make sense. The correct way is not to replace $B_s^{(1)}$ with x , and use the independent increments property to get rid of conditioning:

$$\mathbb{E} \left[\left(B_t^{(1)} - B_s^{(1)} \right)^2 \mid B_s^{(1)} = x \right] = \mathbb{E} \left[\left(B_t^{(1)} - B_s^{(1)} \right)^2 \right] = t - s.$$

- b. • Many students argued that

$$\begin{aligned} \mathbb{P} \left(\left| B_t^{(1)} - B_t^{(2)} \right| \leq r \right) &= \mathbb{P} \left(\left\{ B_t^{(1)} - B_t^{(2)} \leq r \right\} \cup \left\{ B_t^{(2)} - B_t^{(1)} \leq r \right\} \right) \\ &= \mathbb{P} \left(B_t^{(1)} - B_t^{(2)} \leq r \right) + \mathbb{P} \left(B_t^{(2)} - B_t^{(1)} \leq r \right), \end{aligned}$$

where both steps are not correct. In the first step, the union on the RHS of the first line should be changed to intersection because the definition of absolute value gives

$$\left| B_t^{(1)} - B_t^{(2)} \right| \leq r \iff B_t^{(1)} - B_t^{(2)} \leq r \text{ AND } B_t^{(2)} - B_t^{(1)} \leq r.$$

The second step is also wrong (even if we start from the RHS in the first line) because the two events are not disjoint. The correct analysis is:

$$\mathbb{P} \left(\left| B_t^{(1)} - B_t^{(2)} \right| \leq r \right) = \mathbb{P} \left(B_t^{(1)} - B_t^{(2)} \in [-r, r] \right) = 2\mathbb{P} \left(B_t^{(1)} - B_t^{(2)} \in [0, r] \right),$$

where the last step uses the fact that the distribution of $B_t^{(1)} - B_t^{(2)}$ is symmetric around 0 (as we shown above, $B_t^{(1)} - B_t^{(2)} \sim N(0, 2t)$).

- A lot of students use the law of total probability to calculate the probability of the event $\{|B_t^{(1)} - B_t^{(2)}| \leq r\}$, and ends up with a double integral:

$$\begin{aligned} \mathbb{P} \left(\left| B_t^{(1)} - B_t^{(2)} \right| \leq r \right) &= \int_{-\infty}^{\infty} \mathbb{P} \left(\left| B_t^{(1)} - B_t^{(2)} \right| \leq r \mid B_t^{(2)} = x \right) \mathbb{P} \left(B_t^{(2)} \in dx \right) \\ &= \int_{-\infty}^{\infty} \mathbb{P} \left(\left| B_t^{(1)} - x \right| \leq r \mid B_t^{(2)} = x \right) \frac{1}{\sqrt{2\pi t}} \exp \left(-\frac{x^2}{2t} \right) dx \\ &= \int_{-\infty}^{\infty} \mathbb{P} \left(\left| B_t^{(1)} - x \right| \leq r \right) \frac{1}{\sqrt{2\pi t}} \exp \left(-\frac{x^2}{2t} \right) dx \\ &= \int_{-\infty}^{\infty} \left[\int_{x-r}^{x+r} \frac{1}{\sqrt{2\pi t}} \exp \left(-\frac{y^2}{2t} \right) dy \right] \frac{1}{\sqrt{2\pi t}} \exp \left(-\frac{x^2}{2t} \right) dx \\ &= \frac{1}{2\pi t} \int_{-\infty}^{\infty} \left[\int_{x-r}^{x+r} \exp \left(-\frac{x^2 + y^2}{2t} \right) dy \right] dx. \end{aligned}$$

This double integral can be simplified into a single integral via change of variable. To see this, notice that

$$\frac{1}{2\pi t} \int_{-\infty}^{\infty} \left[\int_{x-r}^{x+r} \exp\left(-\frac{x^2+y^2}{2t}\right) dy \right] dx = \frac{1}{2\pi t} \int_{(x,y) \in D} \exp\left(-\frac{x^2+y^2}{2t}\right) dx dy$$

where $D = \{(x, y) : x \in \mathbb{R}, x - r \leq y \leq x + r\}$. Let $u = (x + y)/\sqrt{2}$ and $v = (x - y)/\sqrt{2}$, we can see that $D = \{(u, v) : u \in \mathbb{R}, v \in [-r/\sqrt{2}, r/\sqrt{2}]\}$ and $u^2 + v^2 = x^2 + y^2$. Therefore

$$\begin{aligned} \frac{1}{2\pi t} \int_{(x,y) \in D} \exp\left(-\frac{x^2+y^2}{2t}\right) dx dy &= \frac{1}{2\pi t} \int_{(u,v) \in D} \exp\left(-\frac{u^2+v^2}{2t}\right) du dv \\ &= \frac{1}{2\pi t} \int_{-r/\sqrt{2}}^{r/\sqrt{2}} \left[\int_{-\infty}^{\infty} \exp\left(-\frac{u^2+v^2}{2t}\right) du \right] dv \\ &= \frac{1}{2\pi t} \int_{-r/\sqrt{2}}^{r/\sqrt{2}} \sqrt{2\pi t} \exp\left(-\frac{v^2}{2t}\right) dv \\ &= \sqrt{\frac{1}{2\pi t}} \int_{-r/\sqrt{2}}^{r/\sqrt{2}} \exp\left(-\frac{v^2}{2t}\right) dv \\ &\stackrel{z=\sqrt{2}v}{=} \sqrt{\frac{1}{4\pi t}} \int_{-r}^r \exp\left(-\frac{z^2}{4t}\right) dz \\ &= \sqrt{\frac{1}{\pi t}} \int_0^r \exp\left(-\frac{z^2}{4t}\right) dz. \end{aligned}$$

If your final solution is a double integral that is equivalent to the one presented above, you will lose 1 point for not simplifying it.

Q. 2 (Fruity yogurt in the COVID era). a. We denote the process of customers as $\{N_t\}$ which is a Poisson process with rate $\lambda = 10$. Next, we let the arrival process of customers who helps themselves to the bubble tea to be $\{M_t\}$. By the thinning principle, $\{M_t\}$ is also a Poisson process with rate $0.4 \cdot \lambda$. Then, we know that for every arrival of $\{M_t\}$, there is a probability 0.25 that this person fails to read the notice and drinks the hand sanitizer (we denote the new process as $\{L_t\}$). We use the thinning principle again, and we get that $\{L_t\}$ is a Poisson process with rate $0.25 \cdot 0.4 \cdot \lambda = 1$. Thus, the waiting time T_1 for the first arrival of $\{L_t\}$ follows an exponential distribution with rate $\mu = 0.25 \cdot 0.4 \cdot \lambda = 1$. By using properties of exponential distribution, we have $\mathbb{E}[T_1] = \frac{1}{\mu} = 1$ and $\text{Var}[T_1] = \frac{1}{\mu^2} = 1$.

b. Let T_k be the time of the k -th arrival of L_t , we have

$$\mathbb{E}[T_1 | T_2 = 1] = \int_0^1 t \mathbb{P}(T_1 \in dt | T_2 = 1)$$

Next, we compute the conditional probability $\mathbb{P}(T_1 \in dt|T_2 = 1)$.

$$\begin{aligned}\mathbb{P}(T_1 \in dt|T_2 = 1) &= \frac{\mathbb{P}(T_1 \in dt, T_2 - T_1 \in ds - t)|_{s=1}}{\mathbb{P}(T_2 \in ds)|_{s=1}} \\ &= \frac{\mathbb{P}(T_1 \in dt)\mathbb{P}(T_2 - T_1 \in ds - t)|_{s=1}}{\mathbb{P}(T_2 \in ds)|_{s=1}} \\ &= \frac{e^{-t}dt \cdot e^{-(1-t)}d1}{e^{-1}d1} = 1dt.\end{aligned}$$

Here the second equality is due to the independence property between T_1 and $T_2 - T_1$, and the third inequality follows from the density of exponential distribution (for T_1 and $T_2 - T_1$) and Gamma distribution (for T_2). So we finally have

$$\mathbb{E}[T_1|T_2 = 1] = \int_0^1 tdt = \frac{1}{2}.$$

Grading comments for Q. 2

- a.
 - Most of students did this part correctly. Only a small number of students failed to figure out the distribution of T_1 or the arrival rate of the new process but gave some analysis on how to compute the arrival rate, in this case, I subtracted approximately 4 points.
 - If the students got the expectation or variance wrong and only involved some computation issue, then 2 points were subtracted for each wrong answer.
 - If the students did a correct analysis for the process (figured out its arrival rate) but didn't compute the variance, then 3 points were subtracted.
- b.
 - Some partial credits (3 points) were given if students correctly wrote the formula of $\mathbb{E}[T_1|T_2 = 1]$.
 - In terms of computing the conditional density $f(t_1|t_2)$, if the intuition is almost correct (using the independence property between T_1 and $T_2 - T_1$ and the distribution of T_2 (Gamma distribution)) but the result was wrong due to some computation issue, then (1 ~ 3) credits were subtracted.
 - Some students used the tower rule to compute $\mathbb{E}[T_1|T_2 = 1] = \int_0^1 \mathbb{E}[T_1|T_2 - T_1 = 1 - t, T_1 = t]\mathbb{P}(T_1 \in dt)$. However, this tower rule is not correct. For the conditional expectation, the correct tower rule should be $\mathbb{E}[T_1|T_2 = 1] = \int_0^1 \mathbb{E}[T_1|T_2 - T_1 = 1 - t, T_1 = t]\mathbb{P}(T_1 \in dt|T_2 = 1) = \int_0^1 t\mathbb{P}(T_1 \in dt|T_2 = 1)$.
 - Some students formulated the problem into $\mathbb{E}[T_1|0 < T_1 < 1]$ or $\mathbb{E}[T_1|N_1 = 2]$, both of which are not correct, since the conditions $0 < T_1 < 1$ or $N_1 = 2$ don't have equivalent meaning with the condition $T_2 = 1$.
 - Some students wrote $\mathbb{P}(T_2 = 1)$ in the denominator which is not correct, since T_2 is a continuous random variable. One should write $\mathbb{P}(T_2 \in ds)|_{s=1}$ instead. In this case, 0.5 points were subtracted.

Q. 3 (Rocky Horror Picture Show). *All probabilities are conditioned on $S_0 = 1$ unless mentioned otherwise.*

- a. Let X_i denote the step size of the i -th step. We know that $\mathbb{P}(X_i = -1) = \mathbb{P}(X_i = 1) = \mathbb{P}(X_i = 2) = 1/3$.
Let S_n be the position of the criminologist after the n -th jump with initial position $S_0 = 1$.

$$S_n := S_0 + \sum_{i=1}^n X_i$$

Define the set of values that criminologist falls as $F = \{-1, 3, 4\}$.

Define the boundary hitting time as $T := \min\{n : S_n \in F\}$. We now compute $\mathbb{P}(T \leq 2)$.

Method 1a: Using events

$$\begin{aligned} \mathbb{P}(T \leq 2) &= \mathbb{P}(\{S_0 \in F\} \cup \{S_1 \in F\} \cup \{S_2 \in F\}) \\ &= \mathbb{P}(\emptyset \cup \{X_1 = 2\} \cup \{\{X_1 = 1, X_2 = 1\} \cup \{X_1 = -1, X_2 = -1\} \cup \{X_1 = 1, X_2 = 2\}\}) \\ &= \mathbb{P}(X_1 = 2) + \mathbb{P}(X_1 = 1, X_2 = 1) + \mathbb{P}(X_1 = -1, X_2 = -1) + \mathbb{P}(X_1 = 1, X_2 = 2) \\ &= \mathbb{P}(X_1 = 2) + \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 1) + \mathbb{P}(X_1 = -1)\mathbb{P}(X_2 = -1) + \mathbb{P}(X_1 = 1)\mathbb{P}(X_2 = 2) \\ &= 2/3. \end{aligned}$$

The third equality follows from the fact that the events are disjoint, and the fourth equality comes from independence of the steps.

Method 1b: Using the complement

$$\begin{aligned} \mathbb{P}(T \leq 2) &= 1 - \mathbb{P}(T > 2) = 1 - \mathbb{P}(\{S_0 \notin F\} \cap \{S_1 \notin F\} \cap \{S_2 \notin F\}) \\ &= 1 - \mathbb{P}(\{X_1 = -1, X_2 = 2\} \cup \{X_1 = 1, X_2 = -1\} \cup \{X_1 = -1, X_2 = 1\}) \\ &= 1 - \mathbb{P}(X_1 = -1, X_2 = 2) - \mathbb{P}(X_1 = 1, X_2 = -1) - \mathbb{P}(X_1 = -1, X_2 = 1) \\ &= 2/3, \end{aligned}$$

with similar reasoning as above.

Method 2: Using the Law of Total Probability

Define the time before hitting time as $\tilde{T} = T - 1$.

$$\begin{aligned} \mathbb{P}(T \leq 2 | S_0 = 1) &= \mathbb{P}(\tilde{T} + 1 \leq 2 | S_0 = 1) = \mathbb{P}(\tilde{T} \leq 1 | S_0 = 1) \\ &= \sum_{k=-1}^4 \mathbb{P}(\tilde{T} \leq 1 | S_1 = k, S_0 = 1) \mathbb{P}(S_1 = k | S_0 = 1) \\ &= \mathbb{P}(\tilde{T} \leq 1 | S_1 = 0) \mathbb{P}(S_1 = 0 | S_0 = 1) + \mathbb{P}(\tilde{T} \leq 1 | S_1 = 2) \mathbb{P}(S_1 = 2 | S_0 = 1) + \\ &\quad \mathbb{P}(\tilde{T} \leq 1 | S_1 = 3) \mathbb{P}(S_1 = 3 | S_0 = 1) \\ &= 2/3. \end{aligned}$$

The second line follows from the Law of Total Probability.

- b. We have a random walk (not symmetric) with boundary F . Let N be the number of steps until the criminologist falls off the table.

Define $\mu(i) := \mathbb{E}[N | S_0 = i]$. We want to find $\mu(1)$. For $i \notin F$ (meaning $i \in \{0, 1, 2\}$), we can use first step analysis to obtain

$$\mu(i) = \frac{1}{3}(1 + \mu(i-1)) + \frac{1}{3}(1 + \mu(i+1)) + \frac{1}{3}(1 + \mu(i+2)).$$

And we have the boundary conditions $\mu(i) = 0$ for $i \in F$, that is for $i \in \{-1, 3, 4\}$. Solving this system leads to $\mu(1) = 12/5$.

Grading comments for Q. 3

- a.
 - Many students listed the events of falling on the first or second step correctly, but failed to mention that these sets are disjoint and therefore we can just take the sum.
 - A few students forgot about the event $\{X_1 = 1, X_2 = 2\}$ as a way of falling.
- b.
 - A few students did not solve the system of equations or solved it but did not write the boundary conditions.
 - A few students treated this as a symmetric random walk. This is incorrect because we can take two steps to the right.